GW-k: 3D band-structure benchmark

G_0W_0 fundamental gap (eV) / measured allocated core-hours · 12×12×12 k mesh

Material	aug-SZ2	aug-DZ1	aug-DZ2	aug-TZ1	aug-TZ2	Literature gap [ref]
Diamant C (TZ2: 2 Å)	5.559 / 9.3	5.563 / 98.3	5.611 / 150.3	5.568 / 328.6	5.585 / 1507.8	5.60 / 5.69 [2]; 5.72 [4]
Si	1.131 / 3.7	1.092 / 11.1	1.073 / 18.3	1.157 / 44.9	1.161 / 99.1	1.13 / 1.12 [2]; 1.3 [4]
Ge q4	0.539 / 3.2	0.472 / 10.4	0.558 / 17.4	0.555 / 34.1	0.593 / 58.3	0.63 direct; 0.67 Γ–L [1]
3C-SiC	2.315 / 5.9	2.265 / 37.1	2.267 / 55.6	2.329 / 136.5	2.352 / 459.7	2.42 [1]; 2.42 [4]
AlP	2.282 / 3.1	2.354 / 10.2	2.358 / 17.2	2.464 / 43.0	2.518 / 93.6	2.48 [1]; 2.46 [4]
AlAs q3/q5	2.070 / 3.3	1.943 / 10.2	2.011 / 17.3	2.059 / 305.3	2.093 / 464.3	2.31 [1]; 2.36 [4]
AlAs q3/q23	2.052 / 18.2	1.971 / 61.4	2.025 / 67.8	2.111 / 218.5	2.156 / 272.6	2.31 [1]; 2.36 [4]
AlSb q3/q5	1.662 / 3.1	1.489 / 11.9	1.564 / 17.2	1.605 / 305.3	1.652 / 467.5	1.87 [1]; 1.88 [4]
AlSb q3/q23	1.765 / 10.8	1.585 / 111.7	1.650 / 73.0	1.697 / 468.2	1.768 / 1083.2	1.87 [1]; 1.88 [4]
GaP q21/q5	2.164 / 31.7	2.197 / 79.4	2.175 / 135.4	2.303 / 324.5	2.337 / 901.8	2.30 [1]; 2.37 [4]
GaAs q21/q5	1.108 / 29.0	1.019 / 79.3	1.053 / 126.9	1.099 / 264.9	1.122 / 591.2	1.23 [1]; 1.34 [4]
InP q21/q5	1.188 / 20.7	1.078 / 125.0	1.079 / 218.0	1.156 / 504.7	1.191 / 540.6	1.23 [1]; 1.29 [4]
ZnS Zinkblende q20/q6	3.338 / 29.6	3.215 / 99.6	3.269 / 239.1	3.327 / 275.8	3.343 / 627.6	3.46 [1]; 3.48 [4]
ZnSe Zinkblende q20/q6	2.219 / 25.4	2.115 / 102.5	2.200 / 212.5	2.216 / 378.9	2.274 / 479.6	2.55 [1]; 2.65 [4]
ZnTe Zinkblende q20/q6	2.504 / 23.6	1.949 / 92.4	2.042 / 190.8	2.040 / 372.5	2.097 / 510.7	2.27 [1]; 2.33 [4]
CdS Zinkblende q20/q6	1.910 / 16.9	1.928 / 80.2	1.989 / 116.9	2.038 / 560.6	2.073 / 1575.3	2.15 [1]; 2.06 / 2.06 [2]
CdS Wurtzit q20/q6	2.008 / 228.5	2.008 / 833.6	2.069 / 2851.5	2.125 / 1085.4	2.160 / 1290.2	2.05 / 2.19 [2]; 2.19 [4]
CdSe q20/q6	1.113 / 65.7	1.140 / 89.6	1.237 / 107.1	1.258 / 250.5	1.329 / 316.6	1.52 [1]; 1.71 [4]
CdTe q20/q6	1.500 / 49.5	1.242 / 88.7	1.349 / 101.5	1.339 / 229.1	1.404 / 327.7	1.57 [1]; 1.68 [4]
cBN	6.264 / 12.0	6.285 / 56.3	6.234 / 95.8	6.240 / 183.6	6.308 / 1401.2	6.40 [1]; 6.39 [4]
hBN, Bulk	5.908 / 21.6	5.851 / 69.0	5.899 / 103.2	5.940 / 579.0	5.965 / 776.1	6.17 [4]
BP	2.121 / 3.8	2.032 / 11.6	2.021 / 18.0	2.047 / 42.8	2.076 / 105.8	1.99 / 2.11 [2]; 2.13 [4]

GW-k: 3D band-structure benchmark (continued)

G_0W_0 fundamental gap (eV) / measured allocated core-hours · 12×12×12 k mesh

Material	aug-SZ2	aug-DZ1	aug-DZ2	aug-TZ1	aug-TZ2	Literature gap [ref]
BAs	1.842 / 61.8	1.805 / 125.1	1.824 / 184.6	1.829 / 313.6	1.858 / 451.2	1.91 / 1.83 [2]
AlN Wurtzit	6.476 / 15.8	5.736 / 57.2	5.781 / 88.5	5.770 / 492.4	5.846 / 648.1	5.86 / 5.80 [2]; 6.07 [4]
MgO	7.566 / 86.0	7.803 / 115.1	7.376 / 227.4	7.324 / 392.1	7.355 / 936.1	7.55 [1]; 7.36 / 7.52 [2]; 7.51 [4]
LiF	14.249 / 42.9	13.822 / 65.1	13.678 / 100.5	13.682 / 198.2	13.696 / 779.9	13.49 / 14.27 [2]; 13.74 [4]
LiCl	8.441 / 64.3	8.920 / 86.4	8.864 / 151.4	9.040 / 276.8	9.136 / 452.5	8.64 / 8.71 [2]; 9.21 [4]
NaCl	7.626 / 89.0	8.093 / 125.4	7.926 / 227.2	8.181 / 496.6	8.242 / 915.8	7.80 / 7.92 [2]; 9.27 [4]
Ne	18.612 / 1.6	19.106 / 5.3	19.038 / 8.5	19.443 / 17.6	19.399 / 61.9	—
Ar	12.679 / 0.8	12.691 / 2.6	12.854 / 4.2	13.198 / 10.9	13.137 / 18.4	—
TiO_2 Rutil, indirekt	2.819 / 73.3	3.271 / 199.1	3.282 / 502.2	3.443 / 2795.1	3.397 / 6318.9	3.46 [3] direkt; 3.39 [4]
TiO_2 Anatas	3.113 / 311.7	3.620 / 4334.5	3.639 / 11083.5	3.779 / 3521.7	3.727 / 6615.0	3.73 [3]

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